Topological Void Analysis: A Mathematical Framework for Systematic Technical Innovation Discovery in Knowledge Spaces

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Abstract. Identifying where to innovate in a dense technical domain—such as operating systems or hardware/software co-design—is fundamentally a search problem in a high-dimensional knowledge space. Existing approaches rely on keyword search, citation proximity, or human intuition, none of which formalise the notion of an *unexplored region* that is simultaneously relevant to a target goal and absent from prior art.

We present *Topological Void Analysis* (TVA), a mathematical framework that defines *topological voids* as triads (A, B, C) in a dense-sparse hybrid embedding space. A void requires three conditions: (i) both concepts A and B are semantically cohesive with domain anchor C; (ii) their pairwise similarity falls within a calibrated marginality band—avoiding both obvious combinations and unrelated noise; and (iii) they share a sparse lexical bridge while the geodesic midpoint on the embedding hypersphere is unoccupied.

Applied to ∼140k indexed documents, TVA generates 2,128 invention candidates across 96 targets; 90% survive automated quality filtering, yielding 191 REVISE and 1 APPROVE verdict from four-specialist adversarial review (0.05% end-to-end). Two case studies demonstrate the framework surfaces non-obvious connective tissue rather than merely obvious related pairs.

1 Introduction

Modern software systems, especially at the systems-software layer, evolve through incremental invention: a developer notices a gap between two subsystems, proposes an abstraction to bridge them, and the community iterates toward a patch or patent. The *noticing* step is bottlenecked by human attention and domain breadth. A single engineer cannot simultaneously hold in mind the locking semantics of the Linux scheduler, the memory-ordering guarantees of eBPF JIT output, the ELF relocation model, and the BPF verifier's type system well enough to recognise that an IFUNC-style dispatch contract could unify the latter three.

This paper asks: *can we formalise and automate the discovery of unexplored technical gaps?*

We answer yes, and make the following contributions:

1. A formal definition of a *topological void* (Section 4): a triad (A, B, C) satisfying domain cohesion, calibrated marginality, and sparse lexical bridge conditions in a hybrid dense-sparse embedding space.
2. A *vacancy probing* mechanism (Section 5) based on spherical linear interpolation (SLERP) that rejects pseudo-voids whose midpoint is occupied by existing documents.
3. An *adaptive threshold calibration* procedure (Section 6) that derives domain-specific marginality bounds from corpus statistics, removing manual tuning.

4. A large-scale empirical evaluation (Section 8) with two detailed case studies demonstrating that TVA surfaces non-obvious yet technically grounded innovation candidates.

2 Background and Motivation

2.1 Technical Knowledge as an Embedding Space

Pre-trained embedding models map technical documents into a high-dimensional unit sphere [8]. Points close in cosine distance share semantic content; distant points are unrelated. A *knowledge corpus* $\mathcal{K}$ is a finite set of such points.

Innovation in this view is the act of bridging two concepts A, B that are not yet co-located in $\mathcal{K}$, provided their combination is relevant to a target goal C.

2.2 Geometric Convergence Across Model Scales

A key theoretical underpinning of TVA is the *Platonic Representation Hypothesis* [9]: sufficiently trained models—regardless of architecture, modality, or scale—converge toward a common statistical geometry of their shared training world. Concretely, the pairwise distance structure of a compact embedding model (BGE-M3, 1024D) and a frontier LLM are approximately isometric: if $\cos(\text{dense}(A), \text{dense}(B)) \approx 0$ in BGE-M3 space, the same pair tends to be distant in the LLM's implicit representation space.

This convergence provides the theoretical basis for TVA's design. A topological void identified in BGE-M3 space—a pair (A, B) whose geodesic midpoint is unoccupied—serves as a *statistically reliable proxy* for an underexplored region in the frontier LLM's reasoning space. The compact model acts as an efficient navigator; the LLM supplies the high-resolution generative capability to fill the identified gap. Geometric alignment across scales, formalised via CKA [11], supports this proxy relationship empirically.

2.3 Limitations of Prior Art Search

Conventional patent and prior-art search is keyword-driven or citation-driven [12, 16]. These methods retrieve *known* content; they do not identify *missing* content. Knowledge-graph completion approaches [2] predict missing edges but require a pre-defined relation schema and do not model the continuous "marginality" notion central to patentable non-obviousness.

2.4 The Role of the Domain Anchor

The domain query vector $v_{\text{target}} \in S^{d-1}$ represents the *inventor's intent*—the direction of inquiry rather than a specific document. In our implementation, v_{target} is derived from a natural-language target phrase (e.g., "Optimize Linux synchronization using x86 APIC microarchitecture features") embedded at query time. Enumerating a fixed set of 96 target phrases serves *reproducibility*: it allows systematic coverage of a domain matrix and enables re-running the same targets to compare results. The anchor does not constrain the invention; it orients the search. A practitioner deploying TVA in a new domain need only specify their intent as a target phrase and provide a relevant corpus—no other configuration is required.

2.5 Geometric Intuition: Three Points Span a Plane; a Fourth Points the Direction

Before formalising the framework, it is useful to ground the geometric intuition behind the triad (A, B, C). In high-dimensional embedding space, two points define a vector—a direction encoding a semantic relationship. Three points span a two-dimensional subspace: a semantic plane capturing the relational structure between concepts. A fourth point, predicted by applying the established direction to a third anchor, either exists in the corpus (confirming a known relationship) or is absent (identifying a gap). This absent fourth point is the topological void: the position the logic of the semantic plane implies should exist, but where no document currently resides.

This four-point structure underlies classical vector analogy ("A is to B as C is to D") and motivates the design of conditions C1–C4. The vacancy probe (C4) is precisely the check that the predicted fourth position is unoccupied. Conditions C1–C3 filter for positions that are domain-relevant and non-trivially absent—gaps that are meaningful, not merely sparse.

2.6 The Marginality Principle

Patent law requires that an invention be *non-obvious*: too similar to existing work fails the novelty bar; too dissimilar yields an incoherent or non-enabling disclosure. High-value inventions live in a band of *moderate dissimilarity*—far enough from prior art to be novel, close enough to the domain to be useful. This is the informal basis for our formal marginality condition.

3 Why Standard Approaches Fail

Three retrieval baselines were evaluated before arriving at TVA, each failing in a characteristic way. **Cosine top-k** retrieves relevant but redundant results—all from the same well-covered subsystem, with no notion of a gap. **MMR** [4] improves diversity but provides no occupancy guarantee: ~30% of high-MMR pairs had a corpus neighbour within cosine distance 0.08 of their linear midpoint, making them false voids. **Latent vector arithmetic** [13] ($v_{\text{bridge}} = v_{\text{target}} + (v_A - v_B)$) fails in 1024-dimensional contextualised space: anisotropy [8] makes difference vectors nearly orthogonal to semantic axes, and the curse of dimensionality [1] strips them of directional meaning—67% of bridge vectors escaped the semantic manifold entirely. All three lack an explicit *occupancy check*: they rank or construct candidate points without verifying the region is genuinely unoccupied. This motivates the vacancy probe in TVA.

4 The Topological Void Framework

4.1 Notation

Let $\mathcal{K} = \{k_1, \ldots, k_n\}$ be a corpus of technical documents. Each document $k \in \mathcal{K}$ is mapped by BGE-M3 ($d = 1024$) [5] to a dense unit vector $\text{dense}(k) \in S^{d-1}$ and a sparse token set $\text{sparse}(k) \subset \Sigma^*$ (top-5 lexical weights).

Let $\cos(u, v) = u^\top v$ for unit vectors. Let $\text{Sparse}(k) = \{t \in \text{sparse}(k) : t \notin \mathcal{S}\}$ where $\mathcal{S}$ is a domain stop-word list (high-frequency, low-specificity tokens such as `int`, `define`, `linux`).

4.2 Formal Definition

Definition 1 (Topological Void)**.** *Let $A, B \in \mathcal{K}$ be two corpus documents and let $C = m(\text{dense}(A), \text{dense}(B)) \in S^{d-1}$ (Definition 2) be their synthetic* void midpoint*—the geodesic bridge concept. The pair (A, B) forms a* topological void *with respect to a domain query vector $v_{\text{target}} \in S^{d-1}$ if all of the following hold:*

C1 (Domain Cohesion).

$$\cos(\text{dense}(A), v_{\text{target}}) > \tau_{\text{domain}}$$
$$\cos(\text{dense}(B), v_{\text{target}}) > \tau_{\text{domain}}$$

C2 (Calibrated Marginality).

$$\tau_{\text{low}} \le \cos(\text{dense}(A), \text{dense}(B)) \le \tau_{\text{high}}$$

C3 (Sparse Lexical Bridge).

$$\text{Sparse}(A) \cap \text{Sparse}(B) \neq \emptyset$$

where $\text{Sparse}(k)$ *is the top-5 BGE-M3 sparse-weight set of document k after stop-word removal (as defined in Section 4).*

C4 (Vacancy).

$$\max_{k \in \mathcal{K} \setminus \{A, B\}} \cos(\text{dense}(k), C) < \theta_v$$

where $C = m(\text{dense}(A), \text{dense}(B))$ is the synthetic void midpoint (Definition 2) and θ_v is a vacancy threshold.

Intuitively: A and B both point toward the target domain (C1); are neither trivially similar nor unrelated (C2); share at least one meaningful technical token as a conceptual bridge (C3); and the synthetic void midpoint C is unoccupied by any existing document (C4). The idea to be invented lives in the neighbourhood of C—the gap between two related but unexplored concepts.

4.3 Ranking Voids

Valid pairs (A, B) are ranked by a multi-objective scoring functional $\mathcal{H}(A, B; v_{\text{target}})$ that integrates (i) relevance of the synthetic void midpoint $C = m(A, B)$ to the target domain, (ii) a redundancy penalty against already-selected solutions, and (iii) a non-linear marginality reward centred at the band $[\tau_{\text{low}}, \tau_{\text{high}}]$. The precise functional form and weights are omitted per commercial confidentiality requirements, consistent with industry-track submission guidelines.

5 Vacancy Probing via SLERP

A key failure mode of purely similarity-based approaches is the *false void*: two documents that appear to span an empty region but whose midpoint is, in fact, close to an existing document. Condition C4 addresses this.

Definition 2 (Geodesic Midpoint). *For unit vectors $u, v \in S^{d-1}$ with $\theta = \arccos(u^\top v)$, the geodesic midpoint on S^{d-1} is given by SLERP [15]:*

$$m(u, v) = \text{slerp}(u, v, \tfrac{1}{2}) = \frac{u + v}{\|u + v\|}, \quad \theta \notin \{0, \pi\}$$

with fallback to u for antipodal vectors ($\theta = \pi$).

We use the geodesic midpoint rather than an arbitrary interpolation because it is *equidistant* from u and v on the hypersphere: $\cos(m(u, v), u) = \cos(m(u, v), v)$, guaranteeing that the vacancy test is symmetric with respect to both anchors. For non-antipodal unit vectors this simplifies to normalised linear interpolation; the SLERP formulation handles the degenerate case and connects to the broader literature on geodesic midpoints in curved spaces [15].

Vacancy check (C4) is an $O(n)$ dot-product scan of the corpus matrix against $m(A, B)$, requiring no index rebuild.

6 Adaptive Threshold Calibration

Static thresholds fail across corpora with different density profiles. TVA employs a density-aware calibration layer that derives both τ_{domain} and $[\tau_{\text{low}}, \tau_{\text{high}}]$ on-the-fly from the empirical distribution of the current query's candidate pool.

6.1 Domain Threshold τ_{domain}

The domain cohesion threshold combines a data-driven percentile statistic over the cosine-score distribution of all candidates with a corpus-specific floor. The key property is that τ_{domain} rises in dense regions and falls in sparse ones, preventing both over-permissive selection and empty-result failures. Specific parameterisation is omitted per confidentiality requirements.

6.2 Marginality Band $[\tau_{\text{low}}, \tau_{\text{high}}]$

The marginality band is centred at the empirical mode of the pairwise-similarity distribution among domain-cohesive candidates, with width derived from the spread of that distribution. This operationalises non-obviousness: the band excludes the high-similarity tail (obvious combinations) and the low-similarity tail (unrelated noise) without hard-coded constants. Notably, the calibrated band is itself a *domain characterisation*: it quantifies the pairwise semantic distance at which innovation typically occurs in the target technical space and will differ across domains. Cross-domain comparison of these bands is left as future work. Scale parameters are omitted per confidentiality requirements.

7 System Implementation

We implement TVA in a prototype that ingests a heterogeneous corpus of Linux kernel source (parsed via tree-sitter into function/struct/symbol chunks), hardware architecture manuals, academic papers (PDF-extracted), and patent texts. The corpus contains approximately 140,000 indexed documents after deduplication.

Embedding. We use BGE-M3 [5] in local offline mode for both dense ($d = 1024$) and sparse (top-p lexical weights) representations. All computation runs on CPU. The choice of $d = 1024$ is analytically justified in Appendix A.

Dimensionality selection. While frontier LLMs operate in extremely high-dimensional spaces ($D_{\mathrm{LLM}} \geq 12{,}288$), blindly matching this dimensionality is computationally prohibitive and topologically fragile due to distance concentration [1]. We derived the optimal Vector DB dimensionality from the lens of PRH via SVD on a 10,000-document corpus sample, fitting a power-law decay model $\lambda_i \propto i^{-\alpha}$ to the eigenvalue spectrum. The fit yields $\alpha = 1.069$, $\gamma = \alpha - 1 = 0.069$ ($R^2 = 0.929$), indicating a broad, heterogeneous corpus with high intrinsic dimensionality—driven by the orthogonal syntactic structures spanning C source code, legal patent claims, and hardware specifications. Applying the TVA Dimensionality Theorem (Appendix A) with $N = 148{,}978$, $D_{\mathrm{LLM}} = 12{,}288$, and noise constant $k = 0.001$:

$$D^* = \left[\frac{\gamma \cdot \ln N}{k\,(1 - D_{\mathrm{LLM}}^{-\gamma})} \right]^{1/(\gamma+1)} = 1063$$

The nearest standard dimension is 1024, confirming BGE-M3 ($d = 1024$) is analytically optimal for this corpus. Scaling to $d = 3072$ (`text-embedding-3-large`) would incur 300% computational overhead for only +9.5% theoretical resolution gain, while simultaneously increasing spurious void noise.

Index. Dense vectors are stored in a FAISS flat index [10] for $O(n)$ exact similarity search. Sparse tokens are stored in an SQLite FTS5 inverted index for boolean co-occurrence queries.

Candidate generation. For a given target phrase, we embed the target, retrieve the top-K domain-cohesive candidates (C1), enumerate all $\binom{K}{2}$ pairs, and filter by C2, C3, and C4 in sequence. Surviving pairs are ranked and the top-k returned as voids.

Idea generation. Each void is passed to a frontier LLM with a structured prompt instructing it to propose a technical invention disclosure (TID) that bridges the two void concepts toward the target domain. The LLM is not told the scoring details; it receives only the void description and the target.

8 Evaluation

8.1 Setup

We run TVA over a Cartesian matrix of 12 target domains (Linux scheduler, memory management, file systems, eBPF, virtualisation, networking, power management, device drivers, IRQ handling, CXL memory, PCIe, and security) crossed with 8 x86-hardware feature areas (PEBS, AMX, TDX, CXL, APIC, RAPL, EPT, AVX-512), yielding 96 target specifications. For each target, TVA produces up to 10 void triads, each triggering one LLM call for idea generation.

Quality evaluation. We use a four-stage automated filtering pipeline as a proxy for expert review:

1. *Structural check*: validates JSON completeness and absence of fictional kernel APIs.
2. *Reality check*: iterative critique-and-revise with up to 3 rounds; rejects physically impossible ideas.
3. *Adversarial review*: a four-specialist committee (kernel correctness, novelty/prior-art, strategic value, security/stability) each independently assigns APPROVE / REVISE / REJECT with an integer score.
4. *Deterministic verdict*: fatal-flaw, yellow-card, and majority rules aggregate the four specialist votes.

8.2 Quantitative Results

Table 1 summarises the filtering funnel over 2,128 candidate ideas generated across all 96 targets.

The 191 candidates reaching a REVISE verdict received substantive technical feedback from all four specialists (avg. 5.1 issues per specialist per

Table 1: Quality filtering funnel over 2,128 void-derived invention candidates.

Stage	Surviving	Rate	Notes
Void generation (Forager + LLM)	2,128	100%	96 targets, $\leq$10 voids each
Structural check (Professor)	2,107	99.0%	Rejects missing fields, fictional APIs
Reality check (RC, $\leq$3 rounds)	1,905	89.5%	Iterative critique-and-revise
Adversarial review (Debate Panel)	1,250	58.7%	4 specialists, $\leq$2 revision rounds
↪ REVISE verdict	191	9.0%	Substantive feedback; human review candidates
↪ APPROVE verdict	1	0.05%	Stringent: $\geq$3/4 specialist approval, zero fatal flaws

candidate, avg. specialist score 4.0–5.0/5). Under our deterministic rules, REVISE indicates that the committee found genuine technical merit and requested elaboration—it is a *qualified positive result*, not a rejection. The two case studies below are drawn from this REVISE cohort.

One candidate reached the `majority_approval` threshold (3/4 APPROVE, avg. score 4.0/5, zero fatal flaws, two Debate Panel rounds): *Runtime Density-Adaptive x2APIC Cluster Broadcast for Linux TLB Shootdown*, which proposes per-cluster density-adaptive IPI mode selection in `native_flush_tlb_multi()`. We present REVISE candidates as case studies rather than this APPROVE candidate, because ideas receiving near-unanimous expert agreement are, by construction, less likely to exhibit the non-obvious "connective tissue" that defines the topological void—a point we discuss further in Section 9.

8.3 Rejection Taxonomy

To understand *why* ideas fail, we categorise the fatal flaws in the 888 REJECT verdicts by keyword analysis of specialist feedback. Table 2 shows the distribution.

Table 2: Fatal flaw categories (1,058 rejected candidates; counts exceed 1,058 as multiple specialists may flag per candidate).

Category	Count	%
Locking / concurrency model	1,072	35.5%
Other technical issues	757	25.0%
Prior art / novelty overlap	599	19.8%
Incomplete specification	230	7.6%
Security / side-channel risk	199	6.6%
Kernel ABI / notrace violation	165	5.5%

Table 3: Adversarial approval distribution (extended new-format run, N = 2,128 Maverick, 191 REVISE + 1 APPROVE = 192 outcomes; superset of Table 1).

Approval	Count	%/REVISE	%/Maverick
3/4	2	1.0%	0.09%
2/4	6	3.1%	0.28%
1/4	42	22.0%	1.97%
0/4	142	74.3%	6.67%
$\geq$1	**50**	**26.2%**	**2.35%**
$\geq$2	8	4.2%	0.38%

Table 3 shows the adversarial approval distribution. The 75.5% of REVISE candidates that received zero specialist approvals are filtered from the human-review queue—unanimous rejection signals that no specialist found the core concept defensible. The remaining 24.5% ($\geq$1 APPROVE, 1.70% of all Maverick candidates) represent the system's effective human-review yield: ideas where at least one domain expert identified genuine technical merit despite identifying issues requiring elaboration. This stringent end-to-end conversion rate reflects the *upstream human, downstream AI* philosophy: the value lies not in volume but in surfacing a small set of technically grounded candidates for expert follow-up.

Three observations stand out. First, 90.2% of rejections are triggered by `fatal_flaw_reject` (at least one specialist identifies a blocking technical error), confirming that the committee is per-forming genuine domain-specific reasoning rather than surface-level critique. Second, locking and concurrency errors (35.5%) dominate, consistent with the difficulty of safe kernel synchronisation—exactly the class of error that a Linux maintainer would cite when rejecting a patch. Third, prior-art overlap (19.8%) indicates that a significant fraction of void-derived ideas, while geometrically novel in embedding space, overlap with existing techniques when viewed through a deep-domain lens. This motivates future work on tighter prior-art integration in the vacancy probe.

8.4 Case Study 1: TSX-Advisory MGLRU Rotation Capsules

Target and void. Target: *Optimize Linux memory management using x86 TSX microarchitecture features.* TVA surfaced void #2 from 8 candidate voids: concept A is a paper on *optimistic memory reclamation in lock-free programs*, concept B is a storage-technology selection study (TRaCaR Ratio). Sparse bridge tokens: `memory`, `optimistic`, `reclamation`, `access`. The geodesic midpoint $C = m(A, B)$ had no corpus neighbour within cosine distance 0.09—confirming vacancy.

Idea. The LLM proposed *TSX-advisory single-folio same-lruvec MGLRU rotation capsules*: a fail-closed mechanism that wraps the Linux MGLRU folio rotation fast-path in an RTM (Restricted Transactional Memory) speculation region. If the transaction aborts, the fallback acquires the `lru_lock` normally. The proposal introduces per-folio mutation cookies (seqcount-like odd/even) and a fail-closed locked revalidation step to preserve lruvec consistency.

Expert verdict (REVISE — 3/4 APPROVE), 3 rounds. This is the highest-approval result in our new-format cohort.

- *Kernel Hardliner* (APPROVE): "TSX advisory path is architecturally sound; RTM abort falls back correctly."
- *Prior-Art Shark* (APPROVE): "No direct prior art on TSX-advisory MGLRU rotation; non-obviousness defensible."
- *Intel Strategist* (APPROVE): "Strong x86 TSX differentiation story on Xeon platforms with hardware TSX support."
- *Security Guardian* (REVISE, fatal flaw): "TSX/TAA policy gating not specified concretely enough to guarantee fail-closed behavior under incomplete writer-coverage."

After Round 3 revision: mutation-cookie semantics were formalised (monotonic counter, never odd during active optimistic phase), and the TSX admissibility check was made explicit: RTM is enabled only when `X86_FEATURE_RTM` is present *and* the current kernel TAA mitigation policy permits transactional use. Three of four specialists approved the final draft, making this the strongest positive result in the evaluation cohort.

8.5 Case Study 2: Verifier-Derived Synchronization Contracts

Void. Void #5 in the same run: concept A is `ELF_MACHINE_NAME` (an ELF portability macro), concept B is `addend_may_be_ifunc` (a linker IFUNC relocation predicate). Sparse bridge tokens: `addend_may_be_ifunc`, `elf_machine_name`, `x86_64`. The pairwise cosine similarity (0.64) is at the high end of the calibrated band, indicating the pair is related but non-obvious in the target context.

Idea. Despite the weak, oblique signal, the LLM proposed *Verifier-Derived Synchronization Contract Vectors (SCVs)* for the x86 eBPF JIT: an immutable per-program sidecar table in which the eBPF verifier records each synchronisation site's primitive family, memory order, address class, context mask, and admissibility class; the JIT resolves each site exactly once at load time to an inline template or call-equivalent thunk. The idea ran through three rounds of adversarial revision, refining the SCV schema and clarifying LKMM-equivalence requirements.

Expert verdict (REVISE after 3 rounds — qualified positive). The idea underwent three rounds of adversarial revision (all REVISE, no REJECT; avg. specialist score 5.0/5). Representative feedback and responses:

- *Round 1 — Kernel Hardliner*: "Define the stable-feature admissibility rule in terms of existing `x86_cpufeature` APIs." → Added explicit stable-feature whitelist (TSO baseline, CX8, optional CX16); excluded

RTM/HLE and revocable features.

- *Round 2 — Kernel Hardliner*: "The SCV ABI requires precise versioning and endianness specification." → SCV entry schema extended with `linux_primitive_id`, `width_code`, `memory_order`, `admissibility_class`.
- *Round 3 — Prior-Art Shark*: "Claims overlap with generic verifier fact propagation." → Claims narrowed to the normative primitive-family identifier and the one-time admissibility resolution rule.

Significance. This case demonstrates a key property of TVA: the void pair (`ELF_MACHINE_NAME`, `addend_may_be_ifunc`) has no surface-level connection to BPF synchronisation semantics. A human engineer would not naturally connect these concepts; the LLM used the IFUNC dispatch mechanism as a structural analogy for per-site JIT resolution. The 3-round revision trace confirms that the idea was technically grounded enough to survive sustained adversarial critique and emerge with a stronger, more narrowly-claimed design. This is the "non-obvious connective tissue" that TVA is designed to surface.

9 Discussion

Engineering time savings. A 140,000-document corpus yields approximately $\binom{140000}{2} \approx$ 10 billion candidate concept pairs. The practical difficulty is not merely combinatorial: a human engineer has no principled way to decide *which* pairs to evaluate without first examining them, making exhaustive manual exploration effectively intractable regardless of expertise or time. TVA converts this intractable search into a tractable shortlist: 25,536 automated LLM expert-review calls screen 2,128 candidates and deliver **49 human-review candidates** ($\geq$1/4 specialist approval) requiring approximately **49 person-hours** of focused domain expert review. The authors consider the resulting time saving beyond meaningful quantification—the baseline is not "slower," it is "not systematically possible."

Independent expert evaluation. The 8 candidates achieving $\geq$2/4 adversarial approval (7 REVISE + 1 APPROVE) were subjected to independent domain expert review evaluating technical feasibility, kernel correctness, novelty, and claim quality on a 1–5 scale. Of these, 6/8 (75%) were rated technically sound and worth pursuing, with a mean expert score of 3.5/5. The two remaining candidates require significant additional work: one has incomplete `tid_detail` fields due to revision data loss, and one requires a narrower claim rewrite to address prior-art proximity. This 75% precision at the $\geq$2/4 approval threshold corroborates the adversarial committee's discriminative power and supports TVA's end-to-end effectiveness.

Why does a weak void signal still yield a strong idea? The void conditions define the *search region*, not the idea itself. The LLM fills the region with its own parametric knowledge. A sparse but valid bridge token (here, `addend_may_be_ifunc`) acts as a semantic trigger—analogous to how a human expert reading an unrelated paper suddenly connects it to their domain expertise. TVA formalises the triggering mechanism; the LLM supplies the reasoning.

Calibration sensitivity. The adaptive τ_{domain} calibration reduces manual tuning but introduces dependence on corpus density. For very sparse domains, the percentile-based calibration may set τ_{domain} too low, admitting noisy candidates. The vacancy probe (C4) partially compensates by rejecting occupied midpoints.

On the choice of REVISE as positive evidence. We deliberately present REVISE rather than APPROVE verdicts as case studies. Under our deterministic rules, APPROVE requires at least 3/4 specialist approval with zero fatal flaws from any specialist—a bar calibrated for patent-ready enabling disclosures. REVISE with 3/4 specialist approval and substantive technical feedback represents the system's *intended* output: technically grounded innovation candidates that require domain-expert completion, not autonomous patent generation. A system that routinely produces APPROVE verdicts from LLM specialists alone would be under-calibrated rather than superior—it would fail to catch the locking, prior-art, and security issues that Table 2 shows are pervasive in this domain. The 191 REVISE candidates are the value; the single APPROVE (0.05%) rate is evidence of calibration rigour, not system failure.

Paradoxically, near-unanimous (4/4) APPROVE may indicate an idea that is *too* straightforward: an invention obvious enough that all four independent domain experts agree without reservation is, by definition, unlikely to survive a non-obviousness challenge. This mirrors peer review in top-tier venues, where a paper receiving perfect scores from all reviewers on the first round is either a once-in-a-decade breakthrough or a sign that reviewers did not look closely enough. The most patentable candidates in our cohort—those surviving adversarial review with 2–3/4 approval and substantive specialist feedback—sit precisely in the calibrated marginality band that TVA is designed to surface: not so similar to prior art as to be obvious, not so dissimilar as to be incoherent. Expert disagreement is not a failure mode; it is the geometric signature of a genuine topological void.

Evaluation limitations. Our automated adversarial review pipeline is a proxy for formal human expert evaluation. The 49 shortlisted candidates ($\geq$1/4 approval) are ready for domain-expert review; structured evaluation with a pre-defined rubric, inter-rater reliability measurement, and blinded expert scoring is left for future work.

The revision loop issues a single LLM call per round that addresses all four specialists' feedback simultaneously; this "attend-to-all" strategy can lead to trade-offs where addressing one specialist's concerns inadvertently weakens another's approval. A specialist-decomposed approach—one focused revision per specialist, followed by a merge pass—would likely improve approval rates at the cost of 4–5× more inference per revision round, a compute-efficiency trade-off we leave for future work.

Reproducibility note. Specific hyperparameter values and functional forms for $\mathcal{H}$ and the calibration layer are omitted per proprietary constraints. The four conditions C1–C4, the vacancy probe, and the calibration strategy are described with sufficient precision to reproduce the qualitative behaviour; practitioners can derive corpus-specific values from the described procedures. Guidance on calibration ranges is available upon request from the authors.

Meta-evaluation. This manuscript was iteratively revised through five rounds of the Debate Panel itself—the same adversarial review pipeline described in Section 8. Round 1 triggered a Writing Reviewer REVISE citing "narrative cherry-picking," directly prompting the inclusion of Table 2 and the deterministic verdict formalisation. The Math Reviewer issued REJECT (score 3/10) in Round 3 over a type error in Condition C4 ($\texttt{cos}(k, C)$ vs. $\texttt{cos}(\texttt{dense}(k), C)$), which was corrected in Round 4. By Round 5, the Math Reviewer upgraded to **ACCEPT** (score 8/10) and average specialist score reached 6.2.

This trajectory mirrors exactly the design rationale for `PENDING_HUMAN_REVIEW`: the system did not produce a final APPROVE verdict autonomously—the human author intervened at each round to address specialist feedback. Ideas (and papers) that survive multiple adversarial rounds without full approval are not failures; they are candidates for the "last-mile" refinement that only human expertise can provide. The system identified and corrected weaknesses in its own originating paper across five rounds of substantive, sycophancy-resistant critique. Notably, the committee also penalised this manuscript for withholding implementation details—the same system that identifies innovation gaps was unwilling to fill one about itself. We regard this as evidence of consistent calibration rather than irony.

Scope boundary: static knowledge spaces only. TVA assumes that the knowledge space has a *stable geometry*—that the corpus is a fixed set of prior art against which voids are defined, and that identifying a void does not alter the space. This assumption holds for *Level 1* systems: technical domains where facts are objective, documents are persistent, and discovering an innovation gap does not cause competitors to immediately fill it. The assumption breaks down for *Level 2* dynamic systems—markets, social coordination, competitive strategy—where the knowledge space is *reflexive*: observing a void and acting on it changes the positions of other agents, reshaping the topology. TVA is not designed for Level 2 systems and makes no claim to validity there. Formalising a reflexive extension of TVA is left as an open problem.

Knowledge fragmentation and the locality of voids. TVA identifies voids relative to the boundaries of a local corpus—not against a hypothetical global knowledge space. The commercial and geopolitical fragmentation of knowledge corpora—proprietary datasets, access restrictions, regulatory boundaries—is structurally analogous to allopatric speciation in evolutionary biology. Each isolated corpus develops its own topological structure and void distribution. TVA operating within such a fragment surfaces voids relative to local prior art, not global knowledge. Just as geographically isolated populations evolve distinct adaptations rather than converging to a single optimum, knowledge systems operating on fragmented corpora will surface distinct innovation directions—a form of parallel, domain-specific exploration rather than convergence toward a unified solution space. This is not a limitation of the framework but a reflection of the knowledge landscape it operates in: the topology of what remains to be invented is itself shaped by what each community has chosen to share.

Broader implications: TVA as a mutation engine for multi-agent systems. The TVA framework as presented operates on a static corpus with a single human-defined intent vector. A broader reading of the architecture suggests a different class of system. If multiple TVA-equipped agents are placed in a shared communication network—each operating on overlapping but locally distinct knowledge subsets—the information asymmetry between agents creates natural pressure toward communication and alignment. In this setting, TVA functions not as a batch query tool but as a *mutation engine*: each void-derived proposal is a novel concept injected into the shared knowledge space, and the adversarial review pipeline functions as a selection pressure that retains grounded mutations and discards incoherent ones. The resulting dynamic is structurally analogous to biological evolution operating on a digital knowledge manifold—without any centrally specified loss function or reward signal. Intent emerges from the friction of interaction rather than from an externally imposed objective. We note this not as a claim but as a structural observation about what the architecture implies at scale.

On the limits of monolithic AI systems. The scope boundary noted above points to a deeper architectural constraint. A single model with fixed weights is, by definition, a finite system operating against an infinite dynamic environment (Level 2 reflexive systems, where observation changes the observed). Any finite static approximation of an infinite dynamic process will eventually fail—not due to insufficient parameters, but due to the fundamental incommensurability of finite and infinite. The evolutionary multi-agent framing sidesteps this constraint by replacing a fixed global approximation with an open-ended generative process: the system never claims to have captured the full space, only to be continuously exploring its boundary. This suggests that the path toward robust general intelligence may lie not in scaling single models but in designing the ecological conditions under which intelligence can emerge and self-organise.

Structural correspondence with the Default Mode Network. The operational mode of TVA—searching for unoccupied midpoints between domain-cohesive concept pairs, unconstrained by any specific task objective—bears a structural resemblance to the Default Mode Network (DMN) in biological neural systems [3]. The DMN activates during undirected cognition and is associated with creative insight, cross-domain analogy, and spontaneous novel association—precisely the operations TVA formalises mathematically. This correspondence is architectural rather than causal: both systems allocate processing capacity to boundary exploration rather than task execution, and both produce outputs whose value is not assessable until evaluated by a downstream critic. Whether this structural parallel reflects a deeper invariance in how intelligence explores knowledge boundaries—across carbon and silicon substrates alike—remains an open question.

On the tension between safety and creativity. The TVA framework surfaces a structural tension worth naming explicitly. Safety filtering in embedding space is operationally equivalent to closing voids: a filter that blocks outputs in unexplored regions of the knowledge manifold is, by construction, a filter that prevents the system from entering those regions. Creativity, as TVA defines it, requires precisely the opposite—moving into unoccupied midpoints between domain-cohesive concepts. Whether this tension is fundamental or merely reflects current implementation choices remains an open question. But the observation motivates a reframing of safety: not as a topological barrier applied after void discovery, but as an evolutionary selection pressure embedded within the review pipeline itself—the role the adversarial committee already plays in this system.

10 Future Work

Recursive Topological Expansion. Currently, TVA interpolates strictly within the empirical boundaries of human-authored prior art. A profound avenue for future research is *recursive innovation bootstrapping*. By vectorising and re-ingesting highly-rated, APPROVE-verdict Technical Invention Disclosures back into the active corpus as synthetic prior art, the system can dynamically alter the topology of the knowledge space. In this autoregressive loop, a synthesised TID acts as a newly established structural anchor in the embedding space, creating secondary topological voids between itself and historically distant concepts. This feedback mechanism would transition TVA from a single-step gap-discovery engine into an autonomous "technology tree" generator—iteratively mapping multi-generational frontiers of systems architecture beyond the immediate human horizon.

Downstream implementation. The structured TID format produced by TVA—comprising problem statement, architecture overview, implementation plan, and draft claims—is designed to be directly actionable. A validated TID can serve as a specification prompt for LLM-based coding agents [6], enabling a seamless path from knowledge-gap discovery to prototype implementation. Future work will evaluate end-to-end pipelines from void discovery through automated patch generation and CI-validated compilation.

Human-architect integration. Ideas that exhaust the automated revision budget without reaching APPROVE are routed to a `PENDING_HUMAN_REVIEW` queue. These candidates represent the frontier where LLM reasoning reaches its current limit—precisely the cases most worthy of expert human attention. A structured human-in-the-loop interface that presents the accumulated specialist critique alongside the draft is a natural next step.

Cross-domain generalisation. While this paper instantiates TVA on a Linux kernel and x86 hardware corpus, the framework is domain-agnostic. Any organisation that maintains a large, embeddable technical knowledge base—hardware design documentation, biomedical literature, materials-science patents, automotive software standards, or enterprise architecture repositories—admits the same void formalization. A unified *organisational knowledge graph* that spans multiple engineering disciplines would allow TVA to surface cross-domain voids: ideas at the boundary between, say, a firmware subsystem and a compiler backend that neither team would discover in isolation. Domain adaptation involves two steps: (i) reconfiguring the four specialist roles in the adversarial review committee with domain-appropriate reviewers (e.g., Clinical Researcher and Drug-Safety Expert for biomedical, Materials Physicist and Manufacturing Engineer for materials science), and (ii) re-calibrating the marginality band $[\tau_{\text{low}}, \tau_{\text{high}}]$ from a domain corpus, as the geometric distance at which innovation occurs varies across fields. Both steps are data-driven and require no manual threshold tuning. The rest of the pipeline—void discovery, LLM generation, and revision loop—transfers unchanged. At scale, this positions TVA as an *enterprise-wide innovation radar*—systematically mapping the unexplored territory across an entire organisation's collective technical knowledge, one topological void at a time.

Multi-corpus routing and domain-

specific Vector DBs. The current implementation ingests a heterogeneous corpus into a single shared vector DB. As domain-specific corpora become more widely available, a natural extension is *automatic intent routing*: given an Anchor C, an LLM first identifies the relevant domain DB (or intersection of DBs), then runs TVA within that subspace. Domain-specialised corpora yield higher γ values (faster eigenvalue decay, more concentrated knowledge), sharpening the marginality band and producing more discriminative voids. Since all DBs sharing the same embedding model (BGE-M3, 1024D) occupy the same geometric space, cross-domain void discovery becomes possible by merging corpora—surfacing voids at the boundary between knowledge islands where neither community would discover the gap in isolation. Re-calibration of τ_{domain} and $[\tau_{\text{low}}, \tau_{\text{high}}]$ is required after each merge, as density profiles differ across domains. The single-corpus design adopted here was an intentional simplification enabling end-to-end validation; the routing and merging architecture is a direct next step.

On the asymmetry of cross-domain voids. The framing of cross-domain TVA as surfacing voids "at the boundary between knowledge islands" warrants clarification. In practice, the Anchor C (inventor's intent) is always rooted in a *primary* domain—it expresses a problem to be solved, not a neutral position between fields. A cross-domain void is therefore asymmetric: the *problem* belongs to the primary domain; the *missing bridge concept* is drawn from an adjacent domain. This mirrors cross-domain innovation in practice—Darwin did not pose a question equidistant between biology and geology; he used geological time-scale reasoning to answer a biological question. A fully symmetric "equidistant anchor" is semantically ill-formed for most practical intents. Future work should formalise this asymmetry: TVA with a primary-domain anchor and a secondary-domain candidate pool is a well-posed problem; TVA with an anchor that straddles two domains equally remains an open design question.

The specialist seam problem in cross-domain review. For a cross-domain TID—say, a kernel-compiler co-design proposal—the Kernel Hardliner evaluates kernel correctness and the Compiler Engineer evaluates compiler correctness, but *neither role is responsible for evaluating the interface between the two subsystems*: the ABI contract, data ownership model, and failure propagation across the boundary. This seam is precisely where cross-domain proposals are most likely to fail in practice. A dedicated *Integration Reviewer* or *Boundary Architect* specialist role—one whose evaluation criteria are defined around the interface rather than either component—is a natural extension of the committee model, but the criteria for such a role remain an open design problem specific to each domain pair.

Cross-domain base rate: an unvalidated assumption. The narrative that cross-domain voids are high-value targets is plausible but empirically unvalidated in this work. Prior literature on atypical combinations in patent and citation networks [17] reports a bimodal distribution: most atypical combinations are noise; a small tail of high-impact outliers drives the positive mean. Whether cross-domain TVA voids exhibit the same distribution—or whether the C1–C4 constraints filter toward the high-impact tail—is not known. A minimal pilot is available without additional infrastructure: partition an existing corpus by subsystem (e.g., `mm/` vs. `net/` within the Linux kernel), treat each partition as a "domain," and compare downstream REVISE/APPROVE rates of cross-partition vs. within-partition voids. This pilot requires no additional calibration. We leave this validation to future work.

Dynamic Topological Voids. The present work establishes *static TVA*: the corpus is fixed, and voids are identified relative to a stable geometric snapshot. A natural and substantially harder open problem is *dynamic TVA*: corpora that evolve over time—as new papers are published, patents granted, and codebases updated—produce voids with a lifecycle. A void that exists today may close as the field advances, or widen as adjacent areas diverge. The open questions are: (i) how to assign a *velocity vector* to a void—is it expanding, contracting, or stable? (ii) how to detect *forming voids* before they are fully established, giving inventors a time advantage; and (iii) how to handle the reflexivity problem noted above, where the act of reporting a void accelerates its closure. Dynamic TVA would require temporal embeddings, void-tracking across corpus snapshots, and a revised vacancy probe that accounts for trajectory rather than position alone. We mark this as the primary open research direction.

11 Related Work

Patent and technology forecasting. [12] propose keyword-based patent maps for technology opportunity identification. [16] apply BERT to prior-art search. Neither formalises the notion of an unexplored region or provides a mathematical criterion for non-obviousness.

Knowledge graph completion. TransE [2] and its successors predict missing triples in knowledge graphs. These methods require a pre-defined relation schema and binary (present/absent) labels; TVA operates on continuous similarity and does not require a schema.

Diversity-based retrieval. Maximum Marginal Relevance [4] selects a diverse set of documents relevant to a query by penalising redundancy. TVA uses a similar relevance-novelty trade-off but applies it to the *generation* of new concepts rather than the retrieval of existing ones.

LLMs for code and creativity. Large language models have demonstrated strong performance on code generation [14, 6] and instruction-following [18]. We treat LLMs as black-box reasoners invoked after void discovery; the novelty is the void identification mechanism, not the generation step.

Topological data analysis. Persistent homology [7] identifies topological features (connected components, loops, voids) in point clouds. Our use of "topological void" is inspired by but distinct from TDA: we define voids in semantic embedding space by algebraic conditions on pairwise similarities, not by simplicial complex homology.

Embedding geometry. [8] show that contextualised representations are anisotropic—concentrated in a narrow cone. Our adaptive threshold calibration is designed to be robust to this property; SLERP avoids the geometric distortion that linear interpolation introduces in anisotropic spaces.

12 Conclusion

We have presented Topological Void Analysis, a mathematical framework for systematic technical innovation discovery. By defining topological voids as triads satisfying domain cohesion, calibrated marginality, sparse lexical bridge, and vacancy conditions in a hybrid dense-sparse embedding space, TVA converts the informal notion of "unexplored region" into a decidable predicate.

Applied to a 140k-document corpus of Linux kernel and hardware specifications, TVA generated 2,128 invention candidates, 90% of which survived automated quality filtering and 191 of which engaged four adversarial expert reviewers with substantive technical critique. Two case studies illustrate that TVA surfaces both obvious-gap ideas and non-obvious connective-tissue ideas that neither keyword search nor human browsing would naturally find.

We release the mathematical framework and threshold calibration procedure for reproducibility; implementation details are available upon request.

Ultimately, TVA embodies a simple design principle: *Upstream, we drive AI. Downstream, AI-driven.* The human sets the direction—the domain, the intent, the Anchor C. The pipeline executes autonomously from there: void discovery, generation, adversarial review, revision. The system surfaces the map; human experts navigate it.

References

[1] Kevin Beyer, Jonathan Goldstein, Raghu Ramakrishnan, and Uri Shaft. When is "nearest neighbor" meaningful? In *International Conference on Database Theory (ICDT)*, pages 217–235, 1999.

[2] Antoine Bordes, Nicolas Usunier, Alberto Garcia-Duran, Jason Weston, and Okta-

vian Yakhnenko. Translating embeddings for modeling multi-relational data. In *Advances in Neural Information Processing Systems*, pages 2787–2795, 2013.

[3] Randy L. Buckner, Jessica R. Andrews-Hanna, and Daniel L. Schacter. The brain's default network: Anatomy, function, and relevance to disease. *Annals of the New York Academy of Sciences*, 1124:1–38, 2008.

[4] Jaime Carbonell and Jade Goldstein. The use of MMR, diversity-based reranking for reordering documents and producing summaries. In *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 335–336, 1998.

[5] Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. BGE M3-Embedding: Multi-Lingual, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 2318–2335, 2024.

[6] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harrison Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, et al. Evaluating large language models trained on code. In *arXiv preprint arXiv:2107.03374*, 2021.

[7] Herbert Edelsbrunner, David Letscher, and Afra Zomorodian. Topological persistence and simplification. *Discrete & Computational Geometry*, 28:511–533, 2002.

[8] Kawin Ethayarajh. How contextual are contextualized word representations? Comparing the geometry of BERT, ELMo, and GPT-2 embeddings. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 55–65, 2019.

[9] Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. The platonic representation hypothesis. In *International Conference on Machine Learning (ICML)*, 2024.

[10] Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547, 2021.

[11] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning (ICML)*, pages 3519–3529, 2019.

[12] Sungjoo Lee, Byungun Yoon, and Yongtae Park. An approach to discovering new technology opportunities: Keyword-based patent map approach. In *Technovation*, volume 29, pages 481–497, 2009.

[13] Tomas Mikolov, Wen-tau Yih, and Geoffrey Zweig. Linguistic regularities in continuous space word representations. In *Proceedings of NAACL-HLT*, pages 746–751, 2013.

[14] Erik Nijkamp, Bo Pang, Hiroaki Hayashi, Lifu Tu, Huan Wang, Yingbo Zhou, Silvio Savarese, and Caiming Xiong. CodeGen: An open large language model for code with multi-turn program synthesis. In *The Eleventh International Conference on Learning Representations*, 2023.

[15] Ken Shoemake. Animating rotation with quaternion curves. In *Proceedings of the 12th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH)*, pages 245–254, 1985.

[16] Martin Srebrovic and Imai Yoko. Applying BERT to patent prior art search. In *arXiv preprint arXiv:2101.02105*, 2021.

[17] Brian Uzzi, Satyam Mukherjee, Michael Stringer, and Ben Jones. Atypical combinations and scientific impact. *Science*, 342(6157):468–472, 2013.

[18] Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel

Khashabi, and Hannaneh Hajishirzi. Self-instruct: Aligning language models with self-generated instructions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 13484–13508, 2023.

A Derivation of the TVA Dimensionality Theorem

This appendix provides the full first-principles derivation of the optimal embedding dimension D^* used in Section 7.

A.1 PRH Resolution Formula

The Platonic Representation Hypothesis (PRH) [9] states that sufficiently trained models converge toward a shared statistical geometry. Regardless of frontier LLM size (D_{LLM} dimensions), its knowledge distribution is never uniform. Applying SVD to the embedding space, eigenvalues λ_i follow a **power-law decay**:

$$\lambda_i \propto i^{-\alpha}, \quad \alpha > 1$$

where α is the manifold decay exponent. Specialised domains have larger α; broad heterogeneous corpora have $\alpha \to 1$.

The **topological resolution** of a D-dimensional Vector DB relative to a frontier LLM is the cumulative explained variance:

$$R(D, D_{\text{LLM}}) = \frac{\displaystyle\int_1^D x^{-\alpha}\, dx}{\displaystyle\int_1^{D_{\text{LLM}}} x^{-\alpha}\, dx}$$

Setting $\gamma = \alpha - 1 > 0$ and evaluating:

$$\boxed{R(D, D_{\text{LLM}}) = \frac{1 - D^{-\gamma}}{1 - D_{\text{LLM}}^{-\gamma}}} \tag{1}$$

This is a logarithmic saturation curve: resolution gains are largest at low D and diminish rapidly with further increases.

A.2 Noise Penalty from Finite Data

With finite corpus size N, high-dimensional spaces become extremely sparse. Geodesic midpoints $C = m(A, B)$ computed in a sparse high-dimensional space are likely to land in structurally empty regions—*spurious voids*—not genuine innovation gaps.

Due to the curse of dimensionality [1] and hyperspherical volume expansion, a fixed corpus of N documents becomes increasingly sparse as D grows: the expected distance between any two points converges, and the fraction of the sphere occupied by the corpus vanishes exponentially. Formally, covering the unit sphere S^{d-1} with N balls of radius ϵ requires $N \propto (1/\epsilon)^D$, so the effective neighbourhood radius scales as $\epsilon \propto \exp(-\ln N/D)$, which grows with D for fixed N. As a result, geodesic midpoints computed in high-dimensional sparse spaces are likely to land in structurally empty regions that are empty only due to data scarcity, not genuine innovation gaps (*spurious voids*). We model this sparsity-induced noise as a penalty proportional to $D/\ln N$, and define the **noise penalty**:

$$P(D, N) = k \cdot \frac{D}{\ln N} \tag{2}$$

where k is a noise-tolerance constant calibrated from the adversarial committee's spurious-void rejection rate.

A.3 Optimal Dimension D^*

Combining Equations (1) and (2) into a unified **topological utility function**:

$$\mathcal{U}(D) = \underbrace{\frac{1 - D^{-\gamma}}{1 - D_{\text{LLM}}^{-\gamma}}}_{\text{PRH resolution (signal)}} - \underbrace{k\,\frac{D}{\ln N}}_{\text{spurious voids (noise)}}$$

Setting $\partial\mathcal{U}/\partial D = 0$:

$$\frac{\gamma\, D^{-(\gamma+1)}}{1 - D_{\text{LLM}}^{-\gamma}} = \frac{k}{\ln N}$$

Solving:

$$\boxed{D^* = \left[\frac{\gamma \cdot \ln N}{k\,(1 - D_{\text{LLM}}^{-\gamma})}\right]^{1/(\gamma+1)}} \tag{3}$$

A.4 Corollaries

Corollary 1 (D_{LLM} decoupling). When $\gamma \gtrsim 0.2$, $D_{\mathrm{LLM}}^{-\gamma} \to 0$ as D_{LLM} grows, so $(1 - D_{\mathrm{LLM}}^{-\gamma}) \to 1$ and D^* becomes independent of the frontier LLM's dimensionality. For broad corpora with small γ (e.g. $\gamma = 0.069$), decoupling is partial but D^* still converges to a finite value independent of further LLM scaling.

Corollary 2 (Logarithmic buffer law). N enters D^* only via $\ln N$. A 10× increase in corpus size raises $\ln N$ by only ∼19%, which after the outer root translates to a very modest increase in D^*. The assumption that embedding dimension must scale proportionally with data volume is mathematically unfounded.

Corollary 3 (Empirical validation). For the TVA corpus ($N = 148{,}978$, $\gamma = 0.069$, $k = 0.001$): $D^* = 1063$, nearest standard dimension 1024D. BGE-M3 ($d = 1024$) lies within 4% of the theoretical optimum— confirming the selection post hoc from first principles.

A.5 Sensitivity of D^* to the Noise Constant k

The noise constant k is calibrated from the adversarial committee's spurious-void rejection rate. To assess robustness, Table 4 reports D^* across a two-order-of-magnitude range of k for the TVA corpus ($N = 148{,}978$, $\gamma = 0.069$, $D_{\mathrm{LLM}} = 12{,}288$).

Table 4: D^* sensitivity to noise constant k.

k	D^*	Nearest std. dim
0.0002	4793	3072
0.0005	2034	2048
0.0010	**1063**	**1024** ← chosen
0.0020	556	512
0.0050	236	256
0.0100	123	256
0.0200	65	256

D^* varies substantially with k, confirming that $k = 0.001$ is load-bearing for the 1024D recommendation. The calibration procedure for k introduces a dependence on the downstream review pipeline; Section 9 (Calibration sensitivity) acknowledges this limitation. For practitioners deploying TVA on a new corpus, we recommend computing the *null-model spurious-void density*—the vacancy probe applied to N uniform random unit vectors on S^{d-1}—as a corpus-independent baseline for k calibration, decoupling it from the LLM committee.